4.2 Exponential Smoothing: Aggregate-Daily

2026-04-08

Goal

Apply exponential smoothing to the aggregate-daily Rossmann sales series and evaluate its forecasting performance using a holdout set.

We compare the ETS model to the observed holdout period and report holdout RMSE.

```r
library(tidyverse)
library(forecast)

daily_data <- readr::read_csv(
  "../../03_wrangling/daily_all_clean.csv",
  show_col_types = FALSE
) |>
  dplyr::mutate(
    Date = as.Date(Date)
  ) |>
  dplyr::arrange(Date)

head(daily_data)
```

```
## # A tibble: 6 x 8
##   Date         Sales Customers PromoShare SchoolHolidayShare OpenStoreCount
##   <date>       <dbl>     <dbl>      <dbl>              <dbl>          <dbl>
## 1 2013-01-01   97235     19491          0              1                 17
## 2 2013-01-02 6949829    834075          0              1               1111
## 3 2013-01-03 6347820    761040          0              0.933           1109
## 4 2013-01-04 6638954    782137          0              0.933           1108
## 5 2013-01-05 5951593    687472          0              0.100           1107
## 6 2013-01-06  143904     26387          0              0.100             19
## # i 2 more variables: Weekday <chr>, IsWeekend <lgl>
```

```r
holdout_horizon <- 42
n_total <- nrow(daily_data)
n_train <- n_total - holdout_horizon

train_subset <- daily_data |> dplyr::slice(1:n_train)
holdout_subset <- daily_data |> dplyr::slice((n_train + 1):n_total)

holdout_split <- tibble::tibble(
  n_total = n_total,
  n_train = n_train,
  n_holdout = n_total - n_train,
  train_start = min(train_subset$Date),
  train_end = max(train_subset$Date),
  holdout_start = min(holdout_subset$Date),
  holdout_end = max(holdout_subset$Date)
```

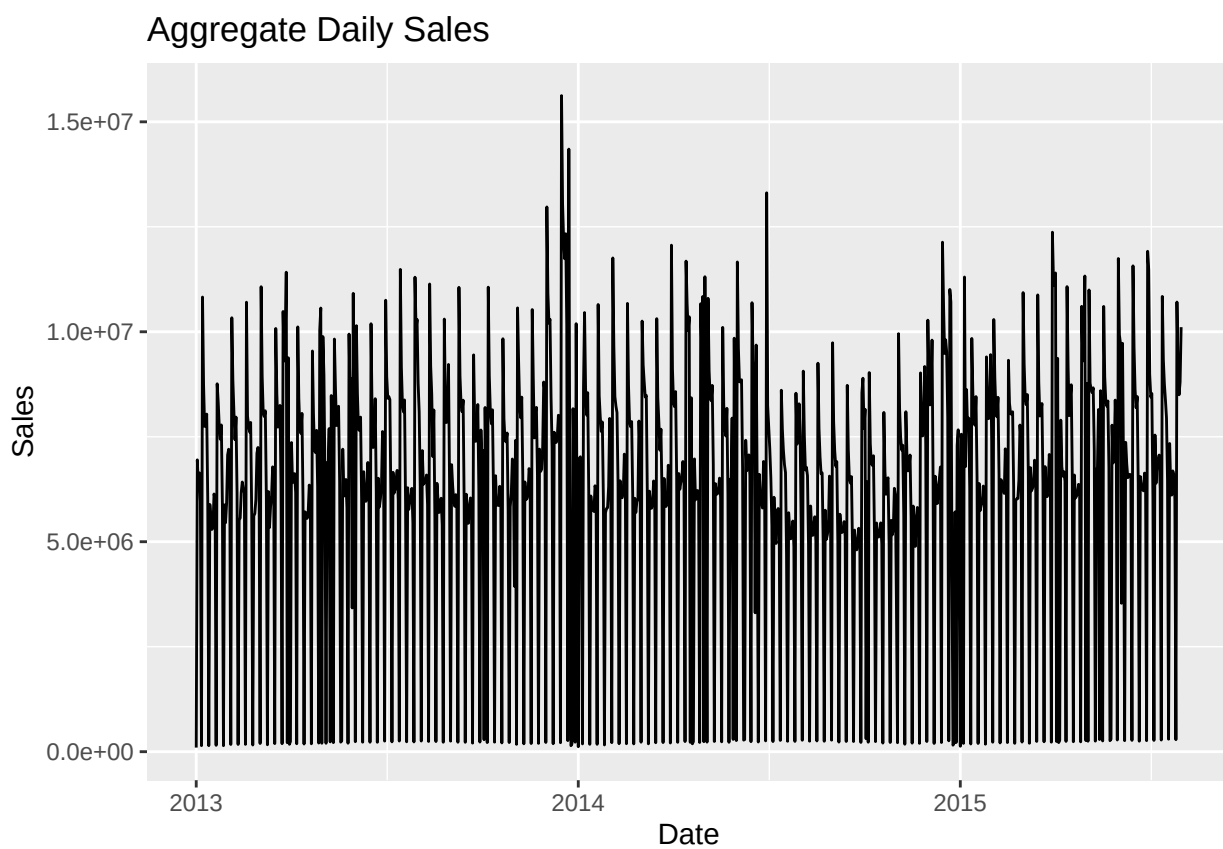
```
)

holdout_split
```

```
## # A tibble: 1 x 7
##   n_total n_train n_holdout train_start train_end  holdout_start holdout_end
##     <int>   <dbl>     <dbl> <date>      <date>     <date>        <date>
## 1     942     900        42 2013-01-01  2015-06-19 2015-06-20    2015-07-31
```

Interpretation

To evaluate forecasting performance fairly, the final 42 observations are reserved as a holdout set, corresponding to a six-week forecasting horizon. The ETS model is fit only on the earlier observations, and its predictions are then compared with the actual sales values in the holdout period. This allows us to assess how well the model would perform in a genuine out-of-sample forecasting setting.

```
ggplot(daily_data, aes(x = Date, y = Sales)) +
  geom_line() +
  labs(
    title = "Aggregate Daily Sales",
    x = "Date",
    y = "Sales"
  )
```



Interpretation

The aggregate daily sales series shows clear repeated fluctuations over time. In particular, a strong weekly pattern is visible, with regular sharp declines on certain days, likely reflecting weekends or days with very

few stores open. This suggests that seasonality is an important feature of the data and supports the use of forecasting methods that can account for recurring weekly structure.

```r
sales_ts <- ts(train_subset$Sales, frequency = 7)
```

```r
ets_model <- ets(sales_ts)

ets_model
```

```
## ETS(M,N,M)
##
## Call:
## ets(y = sales_ts)
##
##   Smoothing parameters:
##     alpha = 0.1284
##     gamma = 1e-04
##
##   Initial states:
##     l = 5820592.519
##     s = 1.4163 0.0338 0.9776 1.1732 1.0903 1.1142
##            1.1946
##
##   sigma:  0.2468
##
##      AIC     AICc      BIC
## 31126.44 31126.68 31174.46
```

```r
summary(ets_model)
```

```
## ETS(M,N,M)
##
## Call:
## ets(y = sales_ts)
##
##   Smoothing parameters:
##     alpha = 0.1284
##     gamma = 1e-04
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##     l = 5820592.519
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##            1.1946
##
##   sigma:  0.2468
##
##      AIC     AICc      BIC
## 31126.44 31126.68 31174.46
##
## Training set error measures:
##                     ME    RMSE     MAE       MPE     MAPE     MASE      ACF1
## Training set -36269.67 1899126 1186058 -101.6393 116.2889 0.567395 0.1705745
```

Interpretation

The selected model is **ETS(M,N,M)**, which means the model uses multiplicative errors, no trend, and multiplicative seasonality. This suggests that the aggregate sales series is mainly characterized by strong seasonal variation rather than a persistent long-term trend. The multiplicative seasonal structure is reasonable here because the size of fluctuations appears to depend on the overall sales level, with larger seasonal swings occurring when sales are higher.

```r
ets_holdout_fc <- forecast(ets_model, h = holdout_horizon)

holdout_predictions <- tibble::tibble(
  Date = holdout_subset$Date,
  actual = holdout_subset$Sales,
  predicted = as.numeric(ets_holdout_fc$mean)
)

head(holdout_predictions)
```

```
## # A tibble: 6 x 3
##   Date        actual predicted
##   <date>       <dbl>     <dbl>
## 1 2015-06-20 6390661  6985649.
## 2 2015-06-21  249849   241550.
## 3 2015-06-22 6552538 10109008.
## 4 2015-06-23 6292372  8533010.
## 5 2015-06-24 6257424  7958452.
## 6 2015-06-25 6211764  7785231.
```

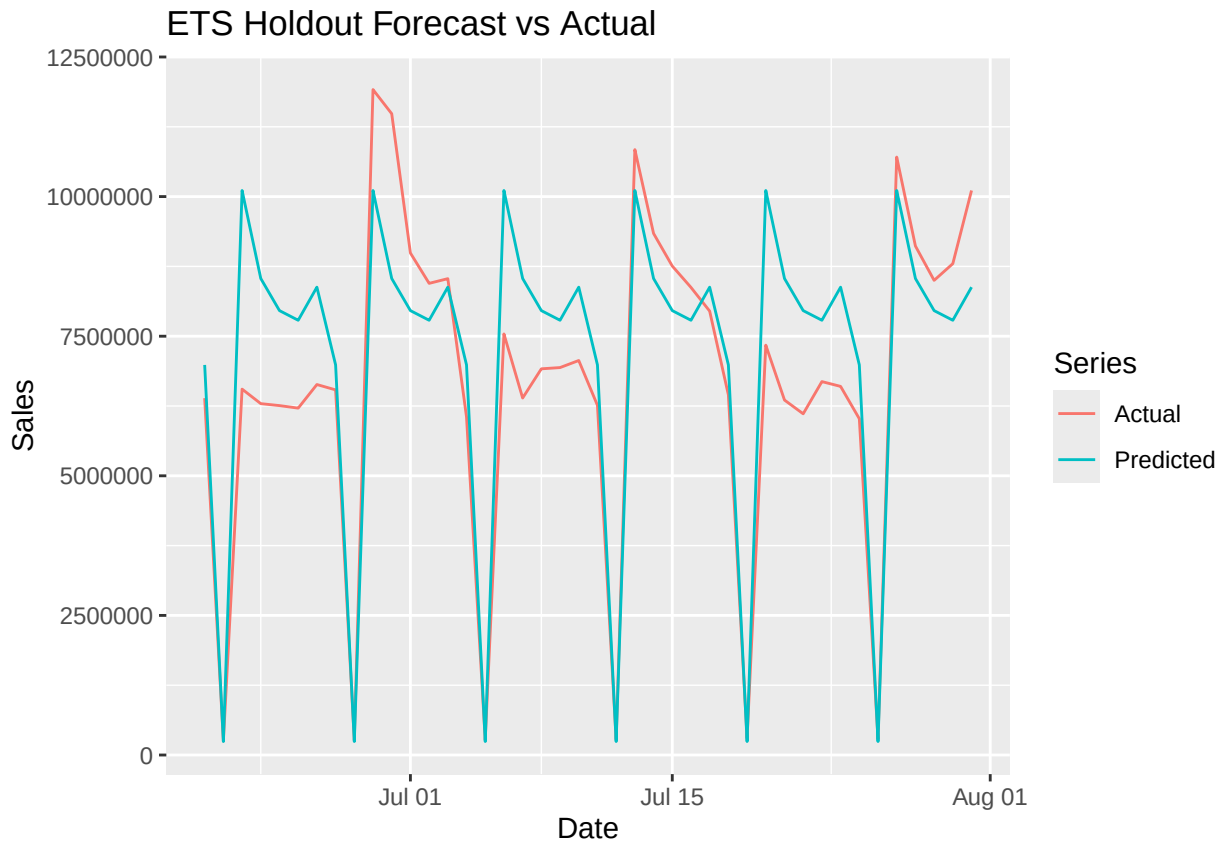
```r
rmse_vec <- function(actual, pred) {
  sqrt(mean((actual - pred)^2, na.rm = TRUE))
}

ets_holdout_rmse <- rmse_vec(
  actual = holdout_predictions$actual,
  pred = holdout_predictions$predicted
)

holdout_rmse_summary <- tibble::tibble(
  model = "ETS",
  holdout_rmse = ets_holdout_rmse,
  n_holdout = nrow(holdout_predictions)
)

holdout_rmse_summary
```

```
## # A tibble: 1 x 3
##   model holdout_rmse n_holdout
##   <chr>        <dbl>     <int>
## 1 ETS       1422687.        42
```

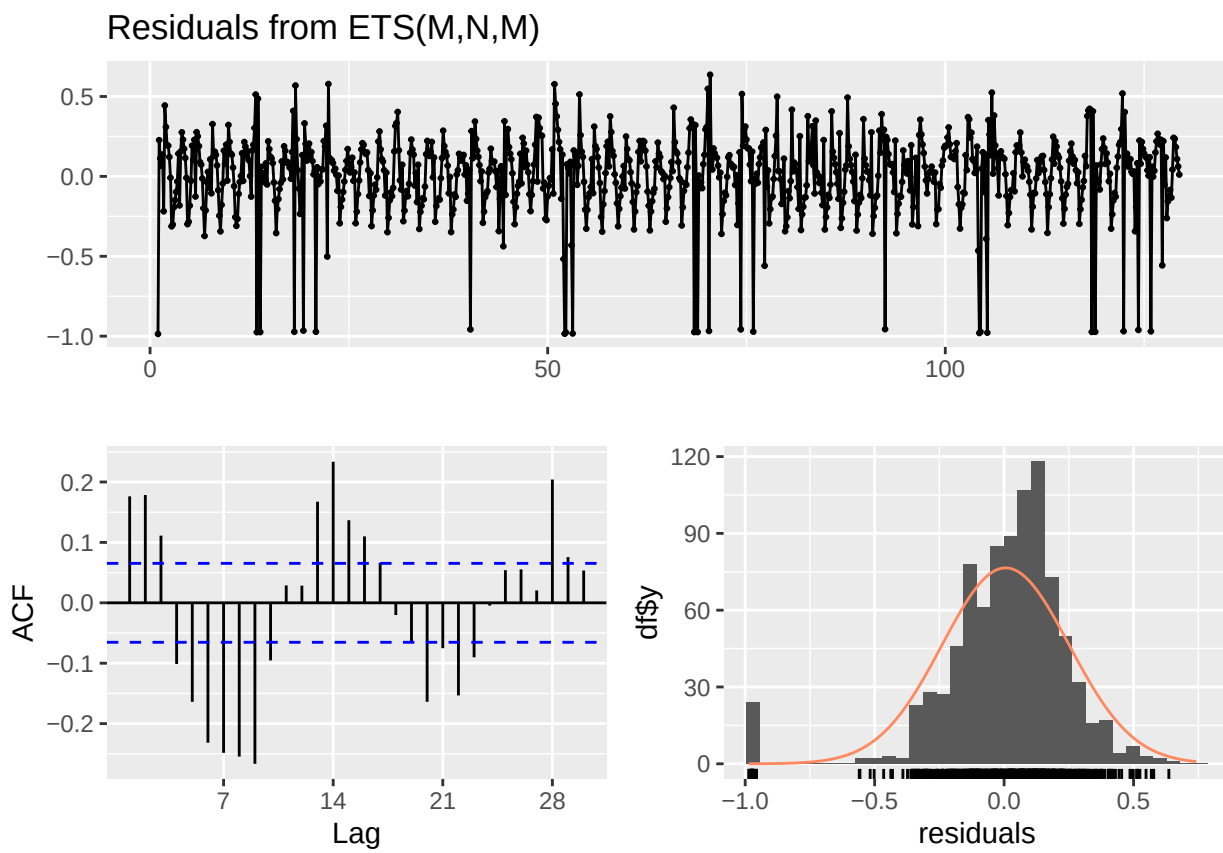
The holdout RMSE for the ETS model is **1,422,687**. This suggests that the model provides a reasonable representation of the weekly sales pattern, although the magnitude of forecast errors is still meaningful and leaves room for improvement through more flexible models.

```r
ggplot(holdout_predictions, aes(x = Date)) +
  geom_line(aes(y = actual, color = "Actual")) +
  geom_line(aes(y = predicted, color = "Predicted")) +
```

```r
  labs(
    title = "ETS Holdout Forecast vs Actual",
    x = "Date",
    y = "Sales",
    color = "Series"
  )
```




```r
checkresiduals(ets_model)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ETS(M,N,M)
## Q* = 415.64, df = 14, p-value < 2.2e-16
##
## Model df: 0.   Total lags used: 14
```

```r
dir.create("results", showWarnings = FALSE, recursive = TRUE)

readr::write_csv(holdout_split, "results/ets_holdout_split.csv")
readr::write_csv(holdout_predictions, "results/ets_holdout_predictions.csv")
readr::write_csv(holdout_rmse_summary, "results/ets_holdout_rmse_summary.csv")
```